Predicting CO_2 Emissions with Machine Learning

Leakage-aware modeling using French EEA vehicle-registration data

PRESENTER

Dr. Ana M. Valencia
Serhat Gemici

MENTOR

Yaniv Benichou

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Presentation objective

From the problem statement to the recommended mode

Core Objectives & Methodological Pillars

Technical Mapping

Relate official WLTP emissions to physical vehicle specifications.

Temporal Validation

Train on 2024 data and evaluate strictly on unseen 2025 registrations.

Leakage Control

Eliminate test proxies to ensure honest and reproducible generalization.

Granular Evaluation

Assess performance across powertrain modes and unseen configurations.

Examine the relationship between technical vehicle specifications and official **Worldwide Harmonized Light Vehicles Test Procedure** (**WLTP**) CO_2 emissions.

Develop on 2024 data and evaluate on a temporal 2025 test set.

Build a reproducible machine-learning workflow that avoids target leakage.

Assess performance beyond an aggregate score: fuel mode and unseen configurations matter.

Data sources and scope

Large, heterogeneous, regulatory vehicle data

From Raw Regulatory Stream to Cleaned Cohort

European Environment Agency **(EEA)**

Agence de l'Environnement et de la Maîtrise de l'Énergie **(ADEME)**

10.78M

Raw EEA 2024 Records

Full European registration registry covering heterogeneous vehicle types.

119k

Cleaned French Subset

Selected for schema consistency and complete technical specification coverage.

Target Variable & Data Selection



- **Target:** `Ewltp`

  Official WLTP CO_2 emissions measured in g/km.

- **EEA vs ADEME Comparison**

  ADEME (2001–2015) suffers from schema drift; EEA French subset was chosen for modeling modern WLTP protocols.

- **2025 Test Cohort**

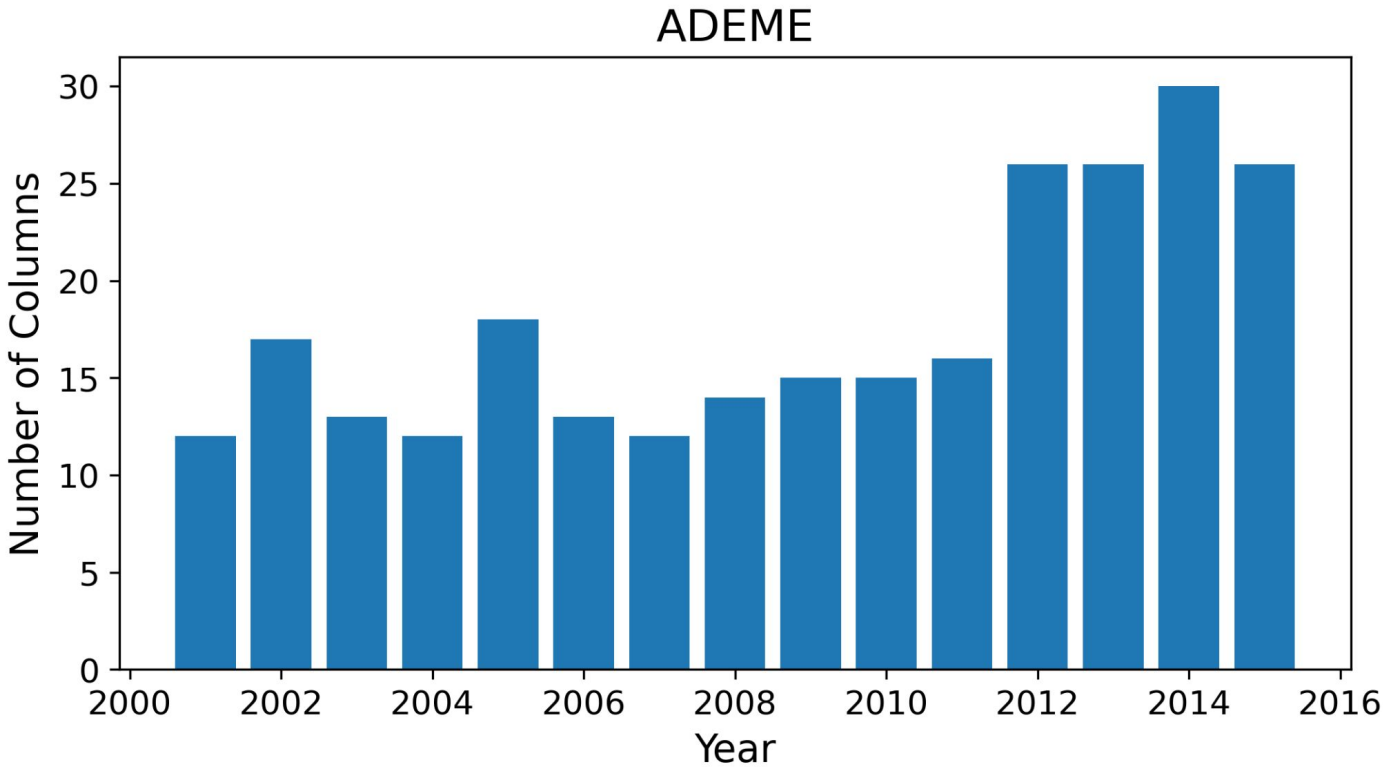
  14,735 records held out completely for temporal assessment.

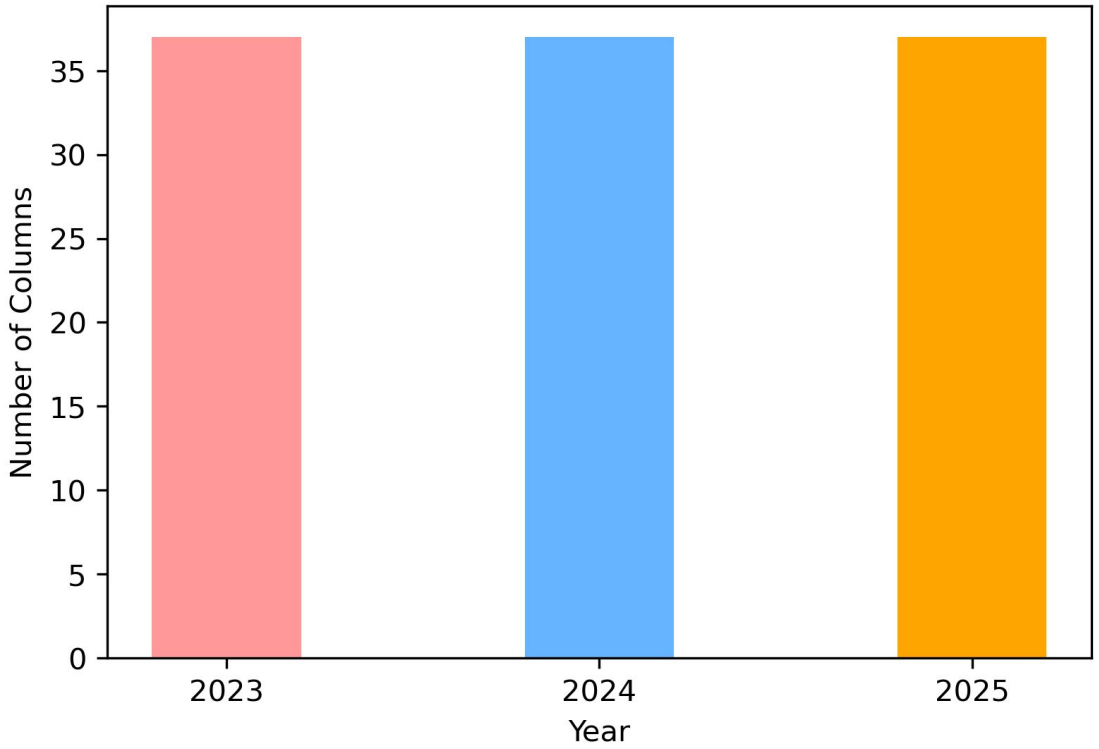
2024
Development data

2025
Temporal test data

Data source exploration

Large, heterogeneous, regulatory vehicle data





ADEME: Three Distinct Schema Phases

- Early volatility (2001–2007)
- Schema expansion (2012)
- Peak and final state (2014–2015)

EEA:

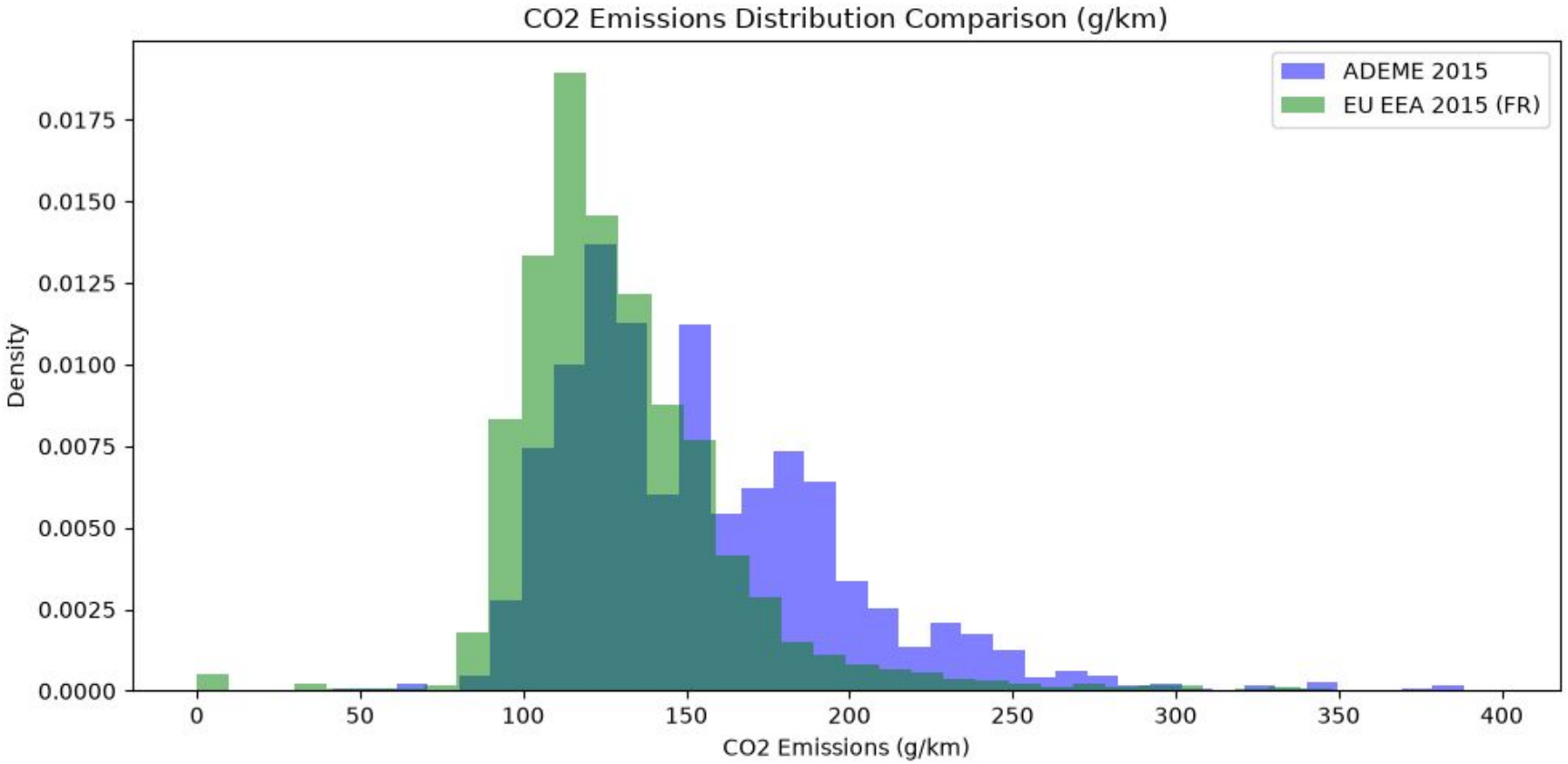
- Constant column count
- Data stability
- Seamless longitudinal analysis

CO2 and pollutant emissions of vehicles marketed in France - data.gouv.fr

Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631

Cross-Dataset parity analysis ADEME vs. EEA 2015 (FR)

Large, heterogeneous, regulatory vehicle data



ADEME dataset features a much heavier right-hand tail, representing higher-emission vehicles ranging from **160g/km to 250g/km.**

EU EEA(FR) dataset drops off rapidly after 150 g/km, concentrating heavily on lower and mid-range emissions.

Both datasets share a common emission band, they clearly represent different vehicle populations or sampling methods

Cross-Dataset parity analysis ADEME vs. EEA 2015 (FR)

Large, heterogeneous, regulatory vehicle data

Why We Selected Only the EEA-Dataset

Market Reality Focus

Sales-weighted registrations reflect actual fleet impact over unweighted catalogs.

Schema Consistency

Identical 37-column structure across 2023-2025 eliminates complex mapping.

WLTP Standarization

Homogeneous emissions testing aligned directly with official EU fleet targets

Bias Elimination

Excludes heavy right-hand tail distortions found in national catalog data

Data quality and preprocessing

A valid model starts with a well-defined analytical problem

Data Preprocessing & Encoding Protocol

1

Column Filtering

Purged zero-variance IDs and missing attributes.

2

Target Encoding

Encoded high-cardinality Maker & Model categories.

3

Domain Imputation

Set electric metrics to zero only when non-applicable.

4

Scaler Hygiene

Fitted scaler strictly on training folds.

Feature selection and target leakage

The prediction task must not reproduce an official test result indirectly

Decision	Features	Technical Justification
KEEP ✓	**m, ep, ec, Ft, Fm, electric range**	Valid physical inputs (mass, power, capacity, fuel type).
REMOVE ✗	**Fuel consumption**	Near-deterministic mathematical proxy for CO_2 emissions.
REMOVE ✗	**Erwltp**	Direct regulatory test value linked closely to target.
CAUTION ⚠	VFN, Tan, T, Va, Ve, RLFI	High-cardinality family codes carrying memorization risk.
REMOVE ✗	**Mt**	Test-procedure specific mass, highly correlated with mass target.

Final Modeling Approaches

Two distinct model formulations based on feature availability throughout the design lifecycle

BASELINE TIER

Model A: Strict Design-Stage

Uses key vehicle characteristics known **before official WLTP test results** are available.

FEATURES INCLUDED:

m (kg) ec (cm3) ep (KW) Ft_diesel Ft_petrol

Ft_petrol/electric Fm_M Fm_P Fm_H

Key Characteristics:

Lowest overall risk of target leakage

Measures ability to generalize to new technical configurations

ENHANCED TIER

Model B: Extended Specifications

Includes additional design and eco-innovation specs assumed available during production planning.

INCLUDES MODEL A + EXTENDED FEATURES:

Electric range (km) IT_28 IT_29 IT_32 IT_33

IT_37 IT_38 IT_none

Assumptions & Usage:

Electric range & eco-innovation technology indicators (IT_*) are assumed ready as design specs prior to predicting official WLTP CO_2 values.

Exploratory findings

The emission distribution separates strongly by powertrain

106.8

Mean Ewltp (g/km)

123

Median Ewltp (g/km)

0-543

Observed Ewltp (g/km) range

-0.786

Range–Emission Correlation

Strong negative Pearson correlation with electric range.

Zero Emission Regime

Battery-electric & hydrogen vehicles cluster near zero tailpipe CO_2.

ICE Dominance

Petrol and diesel dominate higher-emission groups driven by vehicle mass.

Associations

Vehicle mass, engine power and engine capacity are positively associated with emissions.

Associations

Electric range strongly distinguishes PHEV/EV regimes and lower-emission groups.

Experimental design

Moving from optimistic results to stricter validation

Initial Pitfall

$R^2 \sim 0.999$

Early random splits produced overly optimistic benchmarks due to duplicate configurations across train and test sets.

Group-Aware Split

Configuration Grouping

Groups defined by Mass, Capacity, Power, Fuel Type, and Mode. No core-configuration overlap between train and val.

Duplicate elimination

Weighted aggregation

Measured values replaced by registration-weighted means.

2024:113k → 15.9k
2025: 88.9k → 14.7k

Temporal Test

2024 → 2025

2024 data used strictly for development & 5-fold GroupKFold tuning; 2025 dataset held out untouched.

Model comparison

Tree ensembles were strong for this tabular prediction problem

Model Architecture	Val RMSE (g/km)	Val R^2	Role / Status
Tuned ExtraTrees	**6.63**	**0.9838**	**SELECTED**
Tuned Random Forest	7.37	0.9799	Primary Benchmark
Extended ExtraTrees Baseline	7.45	0.9795	Pre-tuning Baseline
MLP (Neural Network)	11.90	—	Deep Learning Benchmark

ExtraTrees: 678 trees; no depth limit; min_samples_leaf = 3.

Search: 20 iterations, 5-fold GroupKFold, using only the 2024 training partition.

Final temporal test results

Strong aggregate performance on the unseen 2025 dataset

5.03*	10.58*	0.9625*	14.7k*
MAE (g/km)	RMSE (g/km)	R^2 Score	2025 Test Records

All 2025 records*

Seen Core Configurations

MAE 3.88 g/km | RMSE 8.69 g/km

Unseen Core Configurations

MAE 11.00 g/km | RMSE 17.29 g/km

The overall score is strong, but generalisation to novel vehicle designs is harder.

Final temporal test results

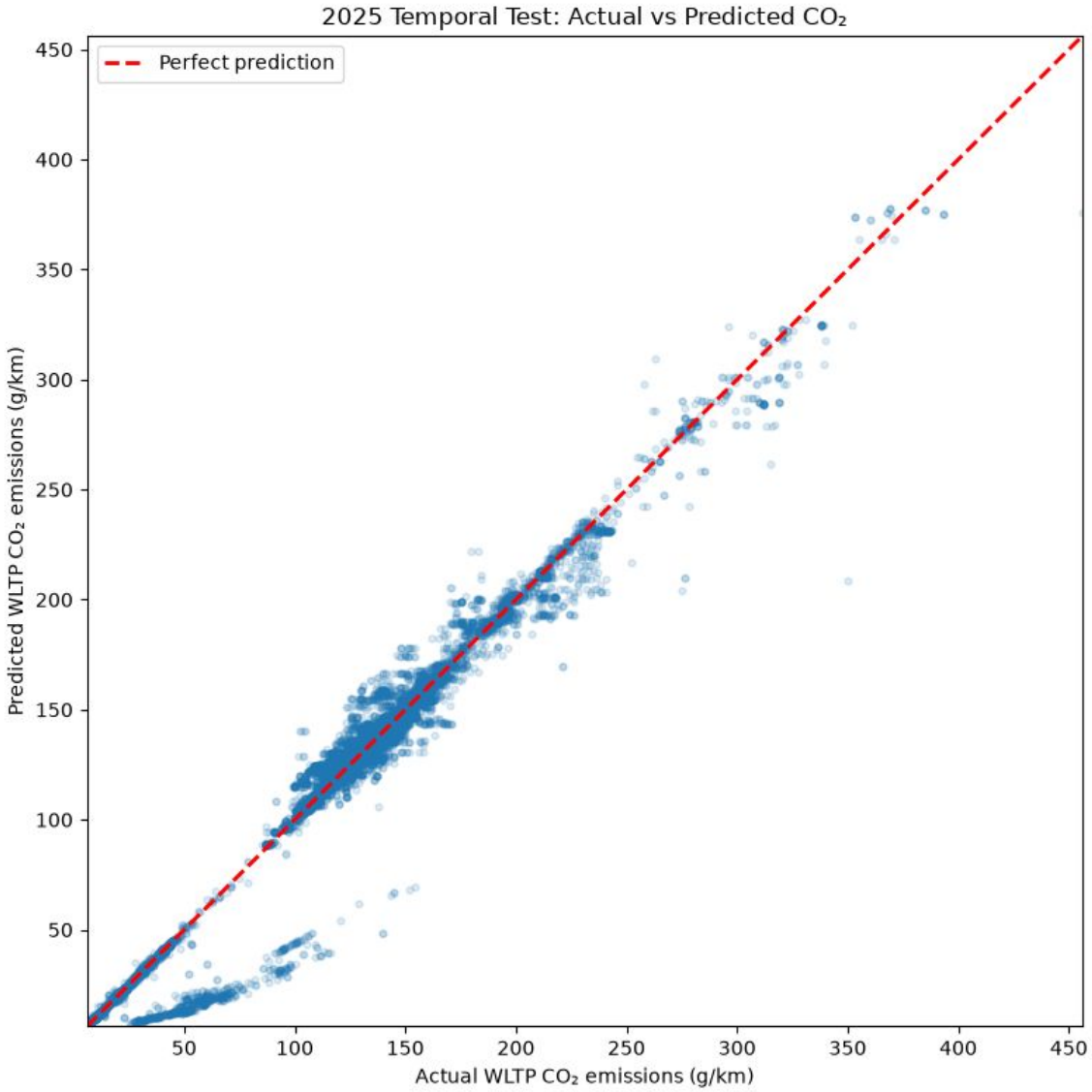
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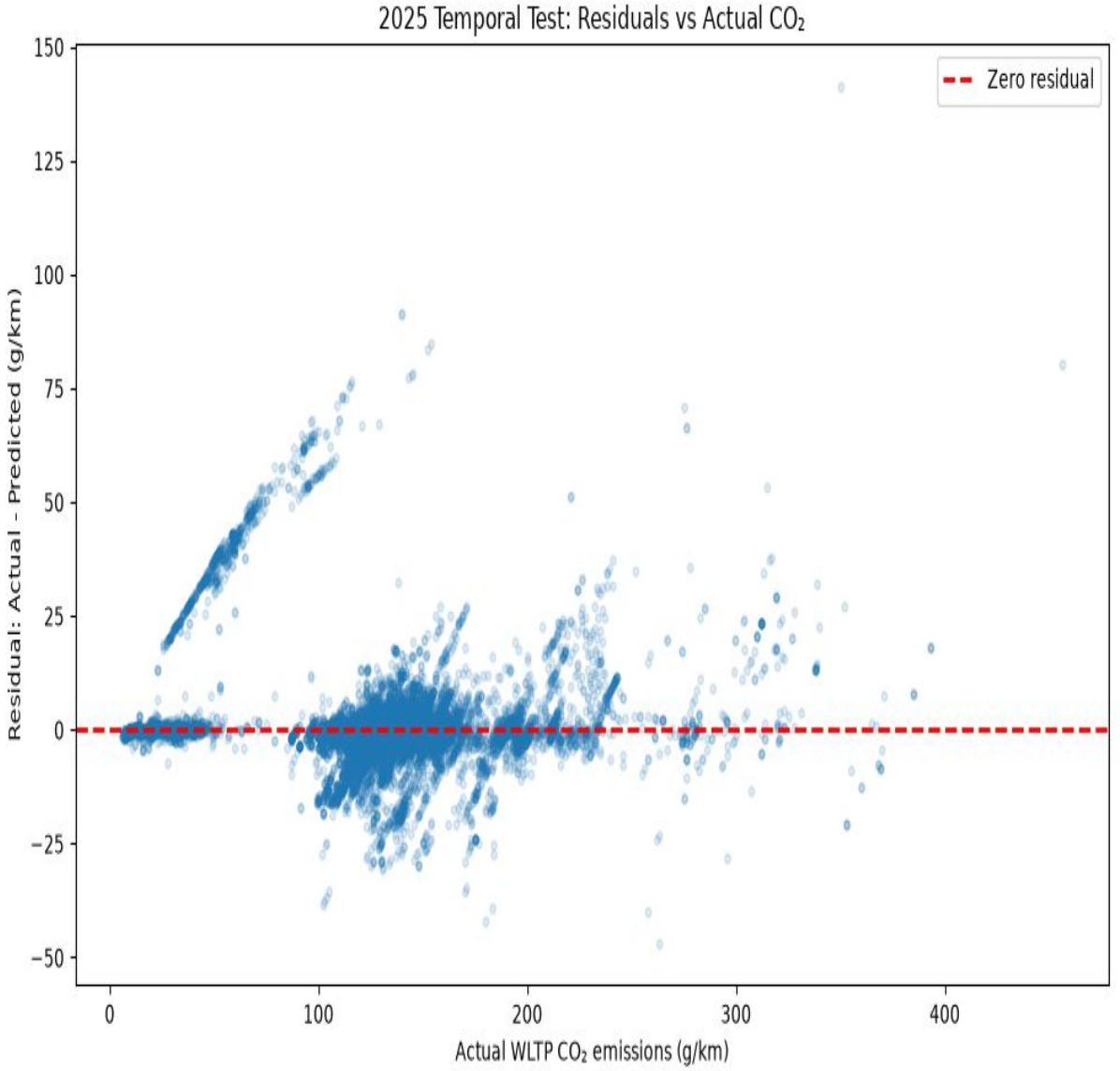
Complete breakdown

2025 evaluation subset	Records	MAE (g/km)	RMSE (g/km)	R^2
All 2025 records	14,735	5.0343	10.5761	0.9625
Seen core configurations	12,337	3.8755	8.6901	0.9733
Unseen core configurations	2,398	10.9959	17.2856	0.9182

Error analysis - Complete Dataset

The structure behind the metrics





Error analysis - Complete Dataset

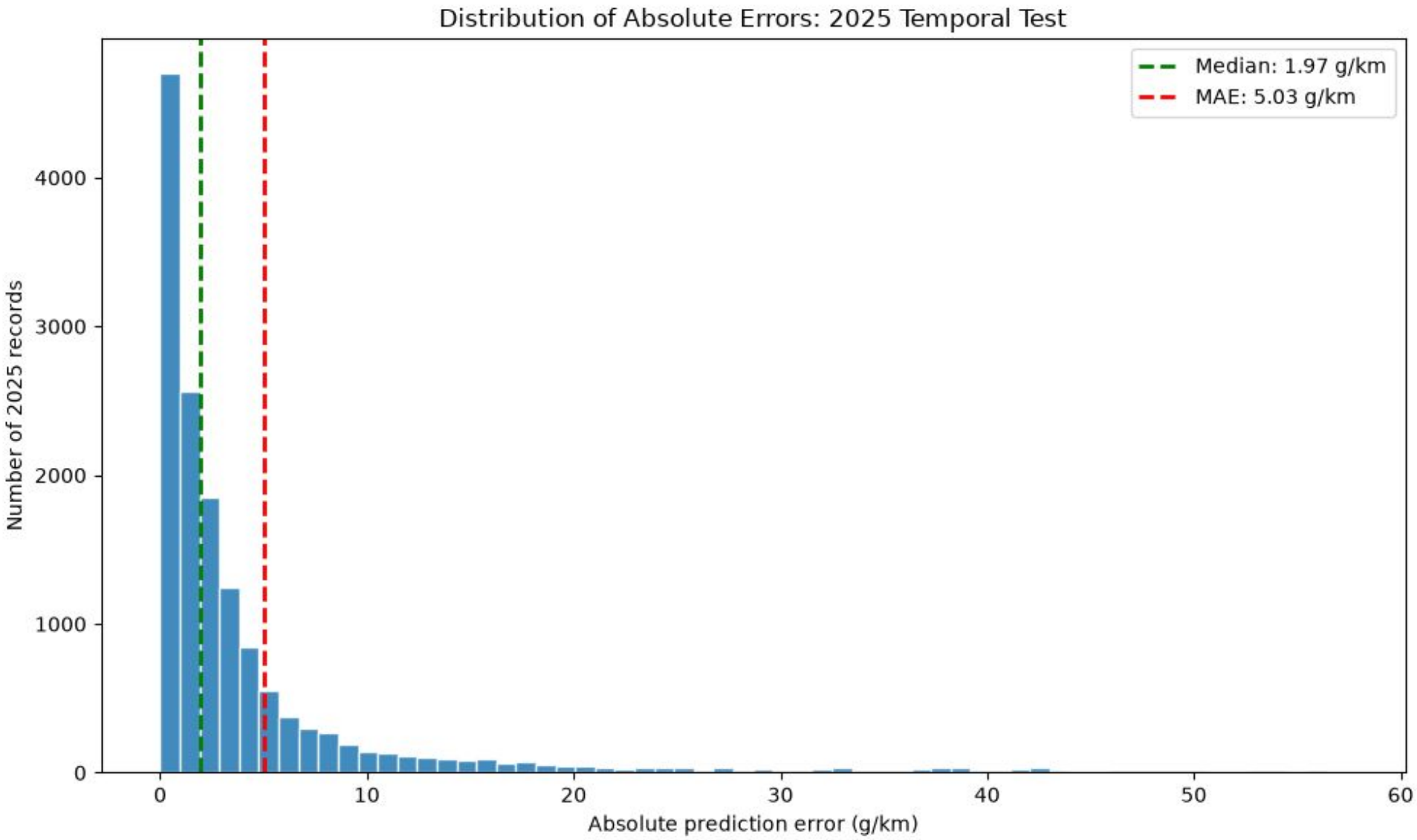
The structure behind the metrics

Largest prediction error:

Feature	Value
Mass	1,985 kg
Engine capacity	6,498 cm^3
Engine power	607 kW
Electric range	13 km
Actual CO_2 emissions	350.0 g/km
Predicted CO_2 emissions	208.88 g/km
Absolute error	141.12 g/km
Core configuration status	Unseen

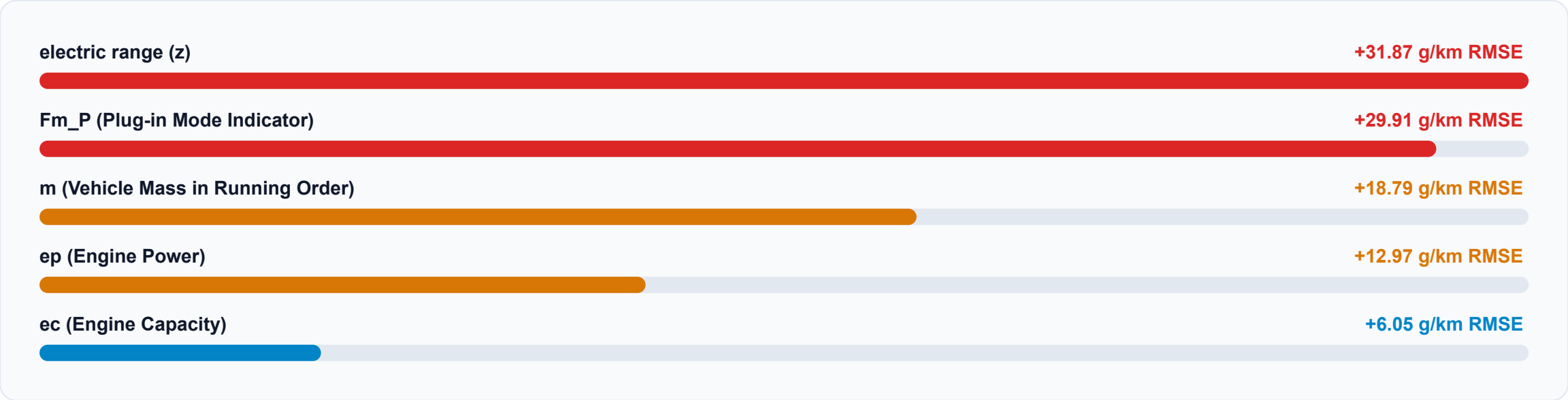
Error analysis - Complete Dataset

The structure behind the metrics



What does the model learn from the most?

Interpretation using permutation importance



Importance is predictive, not causal. Electric range and Fm_P are correlated.*

The model combines physical specifications, powertrain regime and electric-driving information.*

Segment-level results

PHEVs are the main modelling weakness

Fuel mode	All 2025 records	MAE	RMSE	R^2
Mono	6,857	3.85	5.94	0.9746
PHEV	2,695	9.67	20.45	0.5044
Hybrid	5,183	4.92	7.42	0.9430

Mono and Hybrid models generalised consistently.

Unseen PHEVs: MAE 25.07, RMSE 32.77 g/km and R^2 −0.0684.

High-performance PHEVs with short electric range were systematically under-predicted.

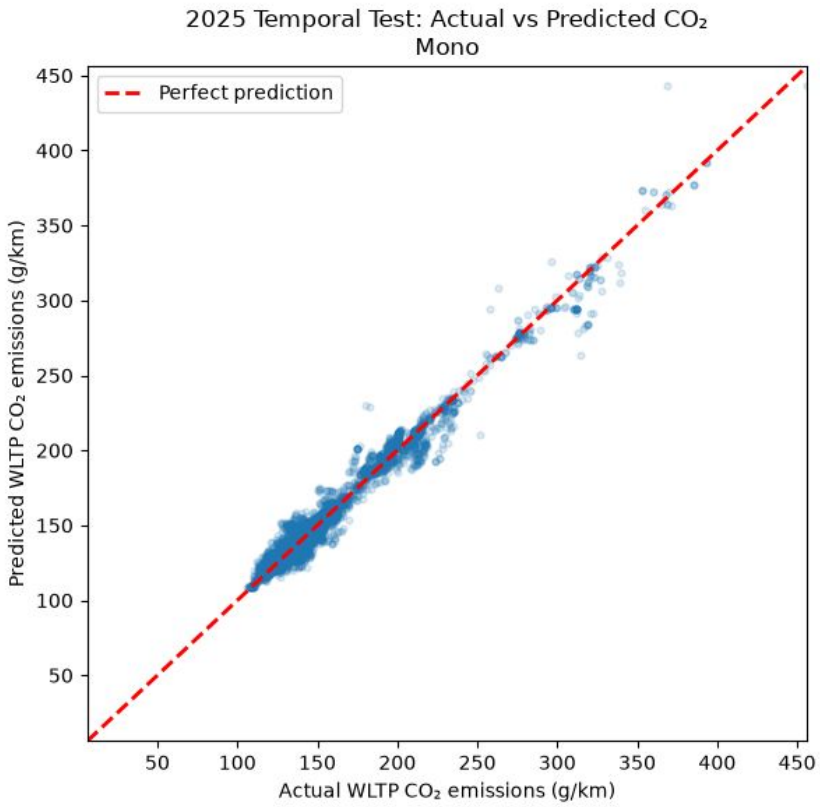
Segment-level results

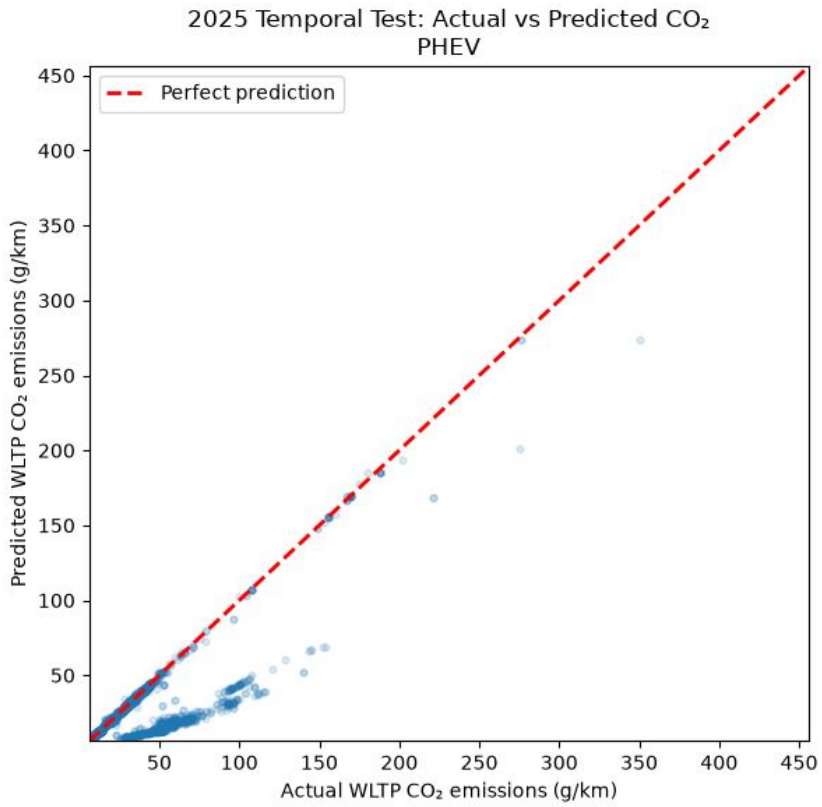
PHEVs are the main modelling weakness

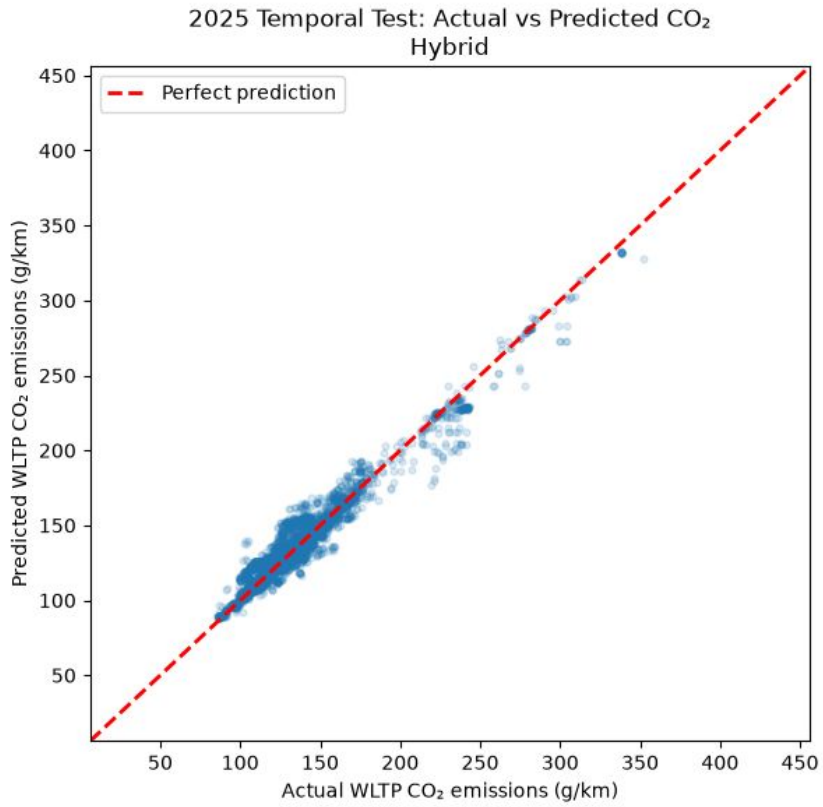
Fuel Mode	Subset	Rows	MAE (g/km)	RMSE (g/km)	R^2
Mono	All 2025 records	6857	3.8459	5.9373	0.9746
	Seen core configurations	5937	3.4254	5.1633	0.9769
	Unseen core configurations	920	6.5589	9.5233	0.9634
PHEV	All 2025 records	2695	9.6732	20.4547	0.5044
	Seen core configurations	2269	6.7831	17.1844	0.6289
	Unseen core configurations	426	25.0665	32.7719	-0.0684
Hybrid	All 2025 records	5183	4.9229	7.4179	0.9430
	Seen core configurations	4131	4.0925	6.3299	0.9541
	Unseen core configurations	1052	8.1839	10.6659	0.9125

Error analysis - Segmented Datasets

The structure behind the metrics

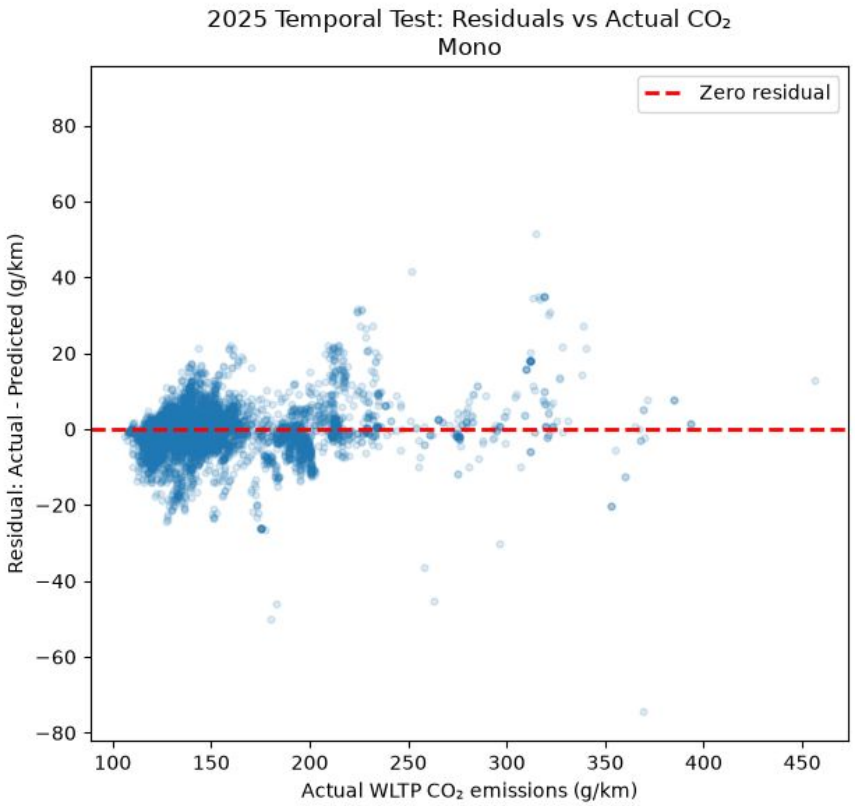


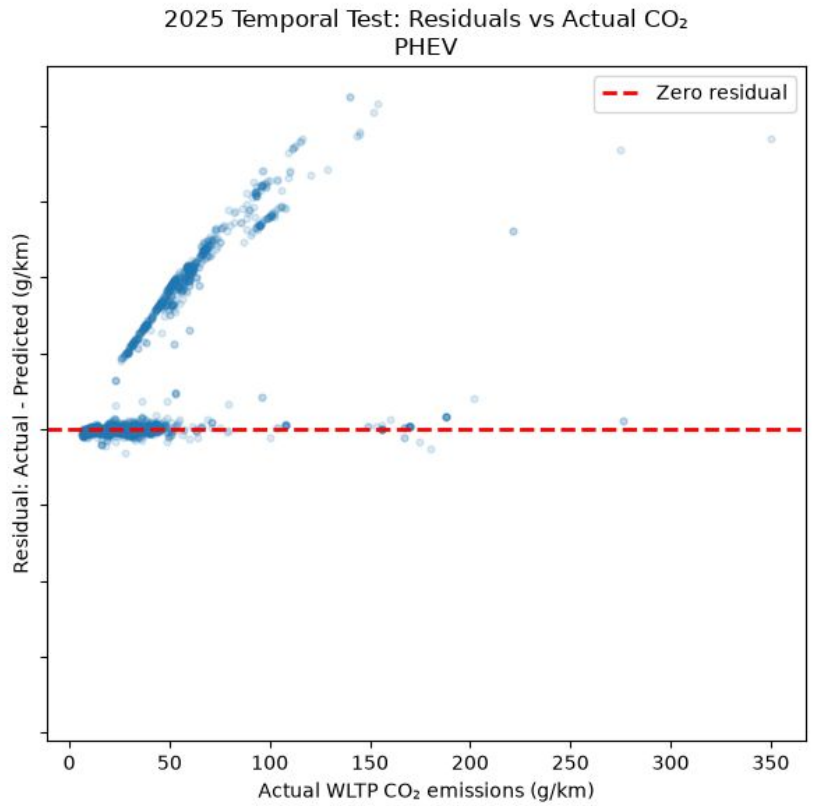


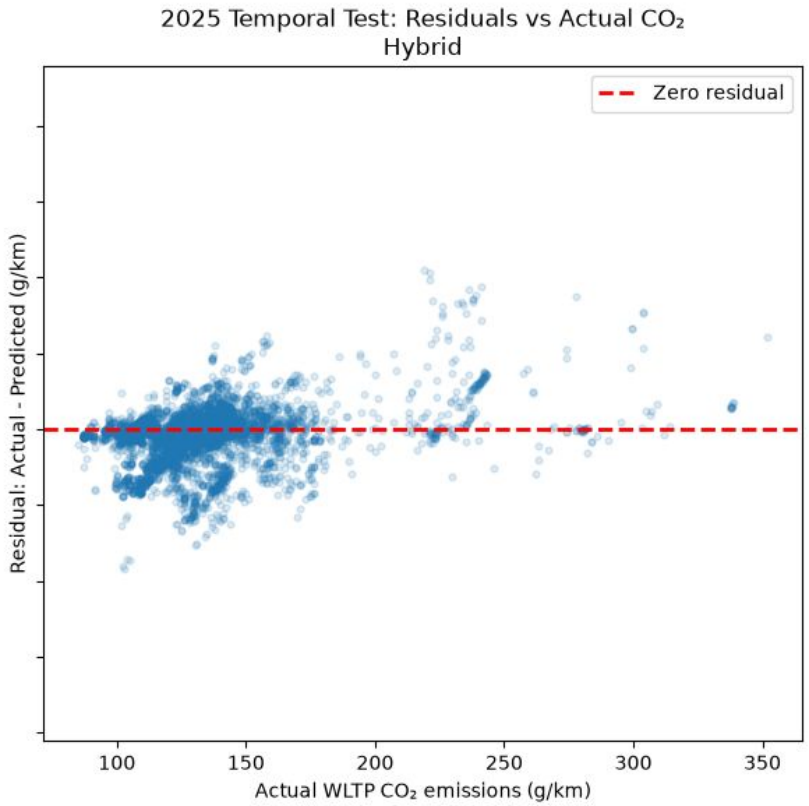


Error analysis - Segmented Datasets

The structure behind the metrics

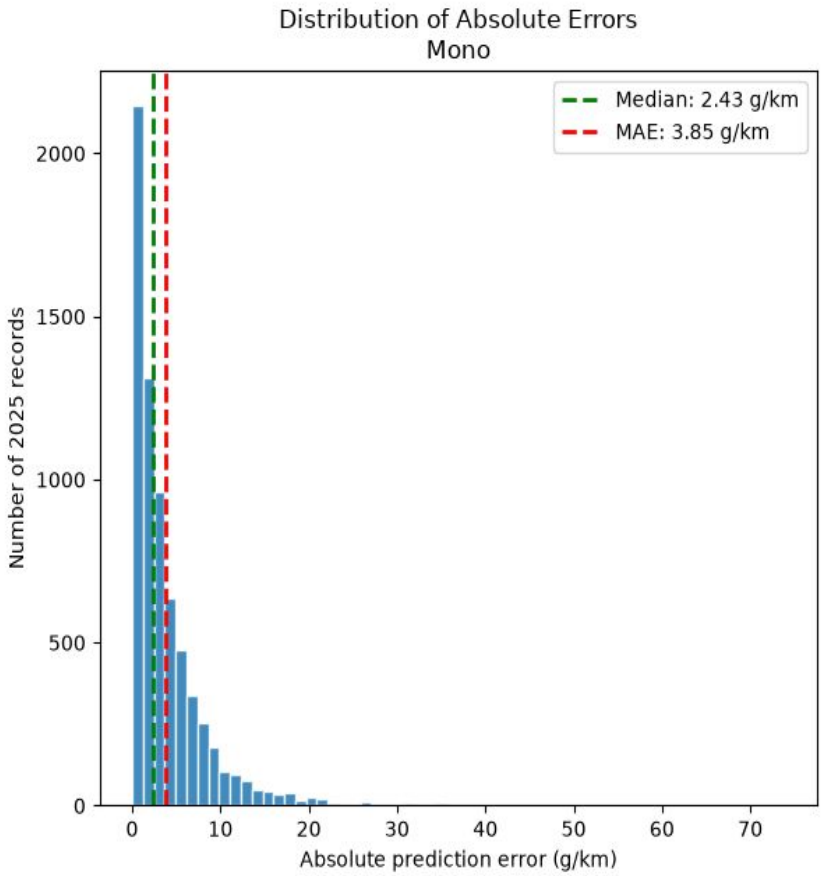


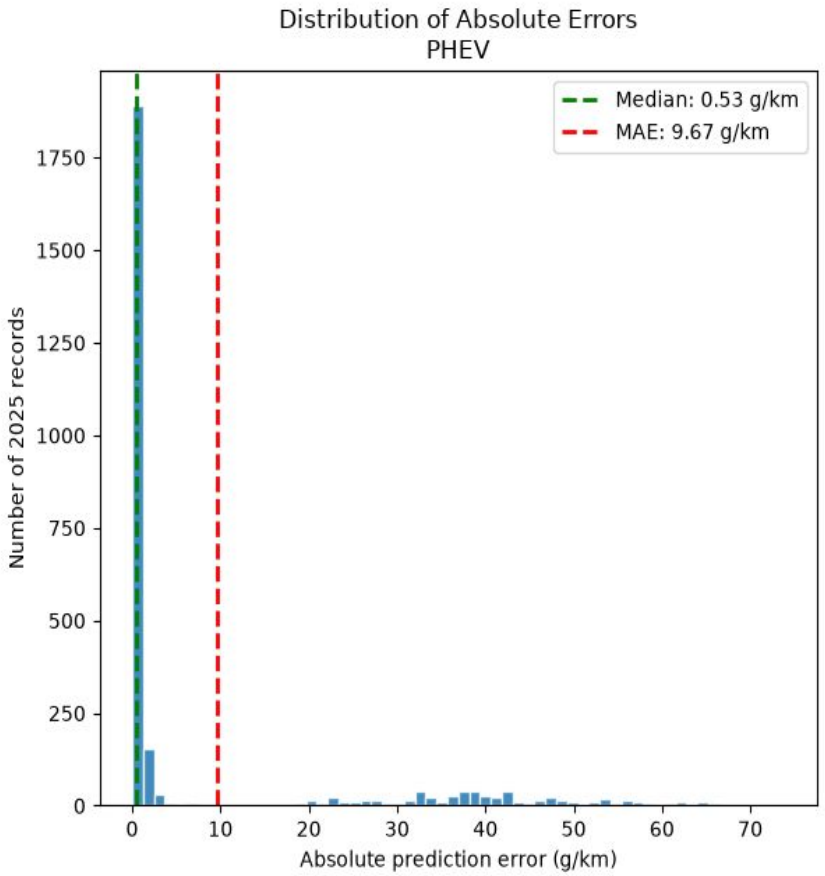


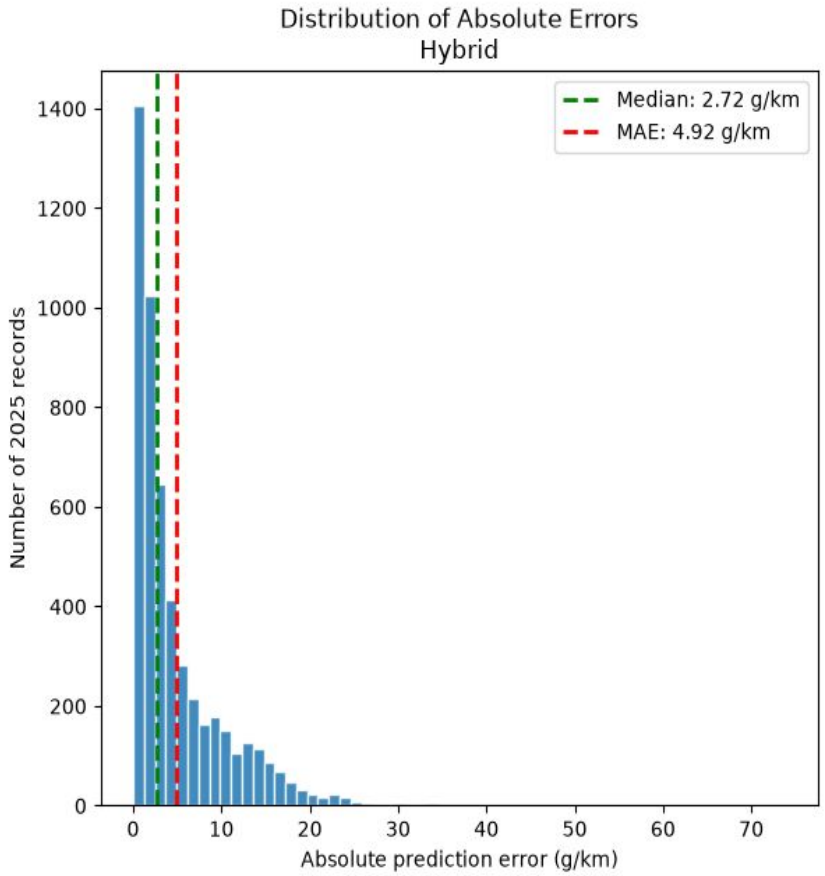


Error analysis - Segmented Datasets

The structure behind the metrics







Scientific and business implications

A useful analytical support tool—not a replacement for homologation

- CO2 emissions can be predicted accurately from technical specifications in this dataset.
- Predictions estimate a laboratory-style WLTP value, not real-driving or life-cycle emissions.
- The model can support early product/fleet analysis, quality checks and scenario comparisons.
- Report results by fuel mode, manufacturer, error tail and seen/unseen status—not only overall metrics.
- Regulatory decisions must rely on official homologation results, not model predictions.

Scientific and business implications

Expert insights

Expert Insights

- The data collection is a pain point; they are recorded very late, hence research data is usually based on projections.
- A project that is based on real data from recent years, and not observations is very valuable.
- A model is more reliable based on observation rather than projection.
- Bias in our data might be cause of fact that the data is reported by the companies themselves.
- Universities collect data on the streets, not in test environments.

Dr. Andrea Orfanoz, Researcher at Max Planck Institute for Meteorology, Hamburg

Recommended use and next steps

A controlled deployment path

1

For early design decisions

Retain the strict feature set containing only pre-test inputs

2

Hierarchical Modeling

Build a dedicated stage-2 model for PHEVs with explicit battery capacity & electric range interaction terms.

3

Dynamic Retraining

Establish annual model updating triggers to absorb new vehicle variant codes as 2026/2027 EEA data releases.

Primary Analytical Tool

Deploy Leakage-Aware ExtraTrees for early fleet planning and quality checks. Disclose error tails by fuel mode rather than relying on global metrics.

Dedicated PHEV Strategy

Develop a two-stage hierarchical model incorporating battery and drivetrain specs specifically for plug-in hybrids.

Demonstration on Streamlit App

Key Takeaway

ExtraTreesRegressor is a strong and practical baseline model; the real challenge is generalising to unseen PHEV configurations.

Thank you

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